


**NATIONAL ECONOMICS UNIVERSITY**

Faculty of Data Science and Artificial Intelligence

**STATISTICS**

# Machine Learning-Based Electricity Consumption Forecasting in Tetouan City

A chronological evaluation of conventional regression models

**GROUP MEMBERS**

1. Dang Nhat Minh — 11247196
2. Giang Hoang Son — 11247222

**Class:** DS66A

**Instructor:** Ph.D. Luong Van Thien

# Can machine learning beat a strong daily pattern?

## PROBLEM MOTIVATION & RESEARCH QUESTIONS

Electricity planning needs the **shape of tomorrow's hourly demand**.

**The practical reference**

If demand repeats every day, copying yesterday is already competitive.

1. How much do conventional models improve on seasonal baselines?
2. Which feature groups add predictive value?
3. How do tuning and the target formulation affect the result?

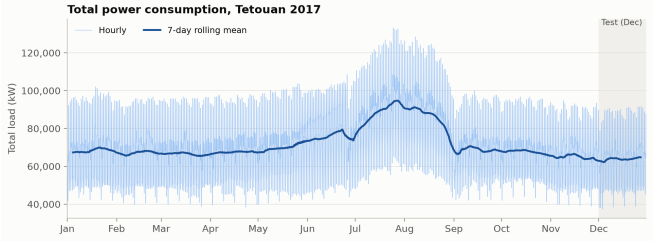
Evaluate future periods and interpret gains against the named baseline.

# The target is mean hourly power across three zones

$$y_t = \frac{1}{6}\sum_{j=0}^{5}\left(Z_{1,t,j} + Z_{2,t,j} + Z_{3,t,j}\right) \quad [\mathrm{kW}]$$

**Z1–Z3 form the target only, never the inputs.**

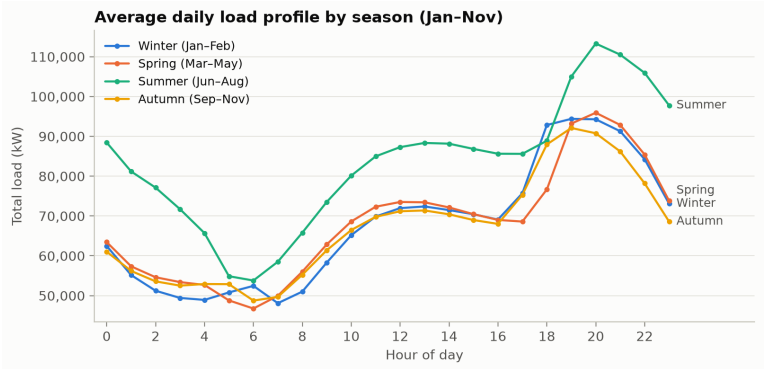
**52,416** × **9**: 10-minute records
timestamp, 5 weather, 3 zone loads
**8,736** hourly means, 2017




Mean power (kW); hourly energy is $y_t \times 1\,\mathrm{h}$ (kWh).

Whole-year view includes December, but no 31 December data; the centred smooth is descriptive.

# Daily shape and season both matter




**JANUARY–NOVEMBER**

**06–07h**
daily minimum

**19–20h**
evening peak

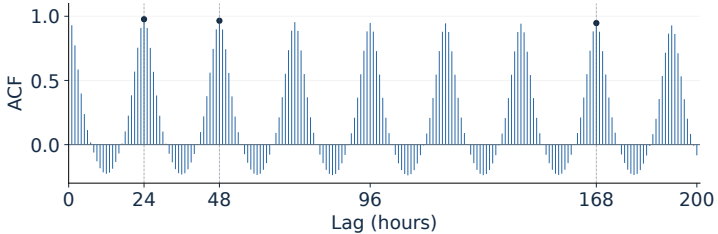
**Summer**
higher load profile

→ **features: `hour`, `hour_sin`, `hour_cos`; temperature for the seasonal level.**

Profiles use January–November development data.

# Today looks like yesterday

**JANUARY–NOVEMBER DATA ONLY**




ACF: **24 h = 0.98**, **48 h = 0.96**, **168 h = 0.95** (weekly peak).

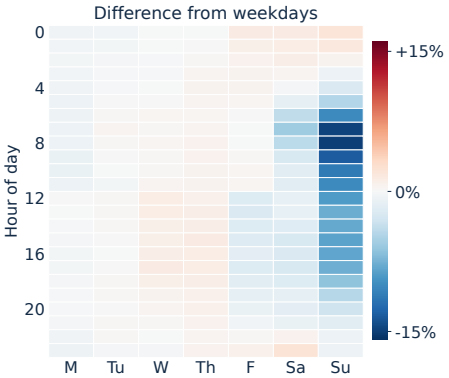
Lag 12 h is negative: day and night are in opposite phases.

Copying yesterday (Naive-24h) is therefore a strong baseline.

→ **features: `lag_24`, `lag_48` (daily history).**

# Sundays and Mondays break the daily pattern

## JANUARY–NOVEMBER / DIFFERENCE FROM MON–FRI AT THE SAME HOUR




Sundays: **5.9% lower** on average; about **15% lower at 7–8 h**.

Saturday mornings: **4–6% lower**.

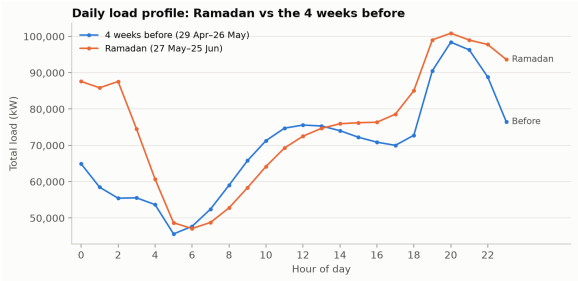
The effect varies by hour; a nonlinear model can learn this interaction.

Copying yesterday mismatches **Monday/Sunday** and **Sunday/Saturday**.

→ **features:** `dayofweek`, `is_weekend`, **cyclic weekday.**

# Holidays and Ramadan move the load

## JANUARY–NOVEMBER DATA ONLY




**Religious holidays** (two Eids, Islamic New Year):
**3–13% lower** than the surrounding days, about $-9\%$ on average.

**Civil holidays**: weaker and uneven, about $-3\%$.
$\Rightarrow$ two separate flags, not one.

**Ramadan**: night load jumps (**+32,000 kW at 02:00**);
temperature differs by only 1.9 °C.

$\rightarrow$ **features:** `is_religious_holiday`, `is_civil_holiday`, `is_ramadan`.

Dates come from the official calendar, known in advance.

# h24 and h1 use different information at issuance

**h24: midnight batch**

00:00 on day $D$: forecast its 24 hourly intervals.

History: $y_{t-24},\ y_{t-48}$.

**Lead to interval start: 0–23 h.**

Lead to interval completion: 1–24 h.

**h1: hourly update**

Update each hour for the next complete hourly interval.

History: $y_{t-1}, y_{t-2}, y_{t-3}, y_{t-24}$.

Uses the latest completed-hour load.

**Prompt measurements assumed**.

The 24-hour lag is a minimum history offset; lead times vary across the daily batch.

Both settings assume prompt load measurements and use observed target-hour weather.

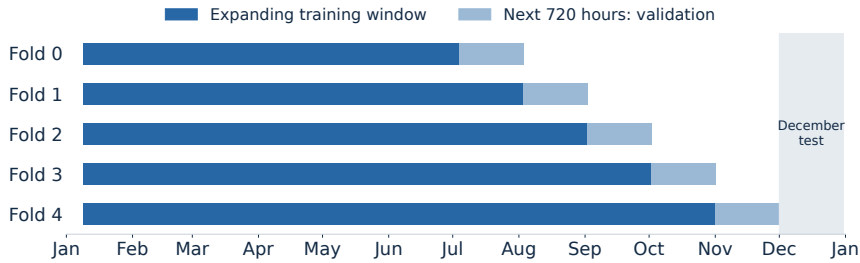
# Seventeen inputs combine timing, weather and history

**MAIN h24 MODEL: FEATURE SET D = 5 ORIGINAL + 12 ENGINEERED**

| Group | Inputs | Built from | Count |
|---|---|---|---|
| **In the original dataset** | | | |
| Weather | Temperature, humidity, wind speed, 2 diffuse-radiation flows | raw columns | 5 |
| **Engineered by us** | | | |
| Calendar | Hour, weekday, weekend flag; sin/cos of hour and weekday | timestamp | 7 |
| Special days | Religious holiday, civil holiday, Ramadan | 2017 calendar | 3 |
| Daily history | $y_{t-24}$, $y_{t-48}$ | past total load | 2 |

# Model selection stays within January–November

**JAN–NOV DEVELOPMENT: 7,825 HOURS** **DECEMBER TEST: 720 HOURS**




After history warm-up, five expanding windows each validate the next 30 days.

Selection: lowest mean CV RMSE → refit on all development data → December.

Tuning and reporting reuse folds (no nested CV); zero gap assumes prompt data.

# Learn the load level—or a correction to yesterday

### TWO SEASONAL BASELINES

$$\widehat{y}_t = y_{t-24} \qquad \widehat{y}_t = y_{t-168}$$

### FIVE REGRESSION FAMILIES

Linear regression / tuned Ridge
Decision tree (DT)
Random forest (RF)
RBF support vector regression (SVR)
XGBoost (XGB)

### TWO TARGET FORMULATIONS

**Absolute:** learn $y_t$.

**Residual ($\Delta$):**

$$r_t = y_t - y_{t-24}$$

$$\widehat{y}_t = y_{t-24} + g(\mathbf{x}_t)$$

Learn the change from yesterday so recent load carries the current level.

**Both target formulations compete on development CV.**

RF averages are range-bounded; boosted trees do not share this strict bound.

# Why predict a correction? July tests extrapolation

**FOLD 0: 4 JULY–2 AUGUST / TUNED MODELS**

**A new load range**

Training maximum: **110,711 kW**
Validation maximum: **133,130 kW**
About **12%** of hours exceed the training maximum.
For $r_t = y_t - y_{t-24}$, **0%** fall outside the training residual range.

**July-fold RMSE (kW)**

| Model | Absolute | Residual |
|---|---|---|
| RF | 5,554 | **3,215** |
| XGBoost | 5,344 | **2,979** |

RF averages cannot extrapolate beyond training targets. Learning a correction reduces the impact of the level shift.

**Residual learning is not necessary for the selected SVR.**

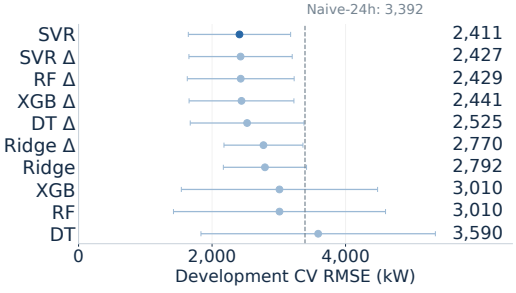
SVR CV: absolute **2,411** vs residual **2,427 kW**; nearly tied.

Target variants were tuned separately; gains also reflect their chosen parameters.

# Absolute SVR leads CV, with close alternatives

Randomized search: **170 configurations** × **5 temporal folds per target type**;

SVR CV RMSE fell **39.3%** versus default SVR.




**2,410.6**

kW mean CV RMSE

± 766.3 kW fold SD.

**Default models:** XGB wins November; LR (**2,792 kW**) wins five-fold CV.

SVR and DT trail Naive-24h (**3,392 kW**) on CV.

**More recent history: h24 vs h1 (CV).** SVR 2,410.6 → 1,866.4 kW ($-22.6\%$).

Same configurations/dates; histories and initial rows differ. CV folds are reused.

# Daily lags drive accuracy; the expanded set hurts

**ABSOLUTE-TARGET RF / FIVE-FOLD CV / FIXED D-TUNED PARAMETERS**

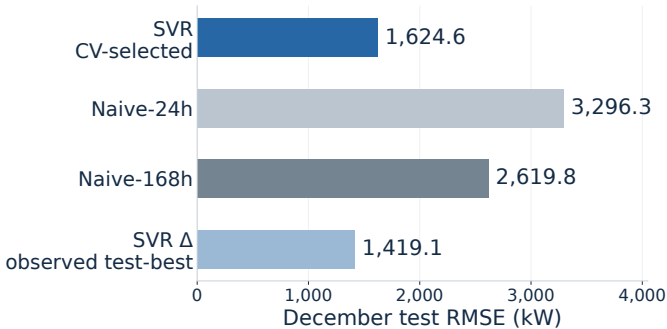
| Set | Feature group | Inputs | RMSE (kW) | Change |
|---|---|---|---|---|
| A | Weather only | 5 | 15,010 | — |
| B | + Calendar | 12 | 10,041 | $-33.1\%$ |
| C | + Special days | 15 | 9,168 | $-8.7\%$ |
| **D** | **+ Daily lags (24 h, 48 h)** | **17** | **3,010** | $-67.2\%$ |
| E | + Weekly lag and rolling summaries | 21 | 3,700 | $+22.9\%$ |

**D is selected:** daily lags give the largest gain; weekly/rolling features worsen RF and XGBoost (XGB **3,010** $\rightarrow$ **4,921 kW**), so they are **excluded from the final model**.

**Set F** = E without weather: RMSE $+1.6\%$ (RF), $+1.9\%$ (XGB), at most $+5.3\%$ (RF $\Delta$).

F is not a test of the selected SVR.

# The selected SVR halves daily-baseline December RMSE




**50.7%**

RMSE reduction vs Naive-24h

**38.0%** vs Naive-168h

MAE: **1,094.6 kW**

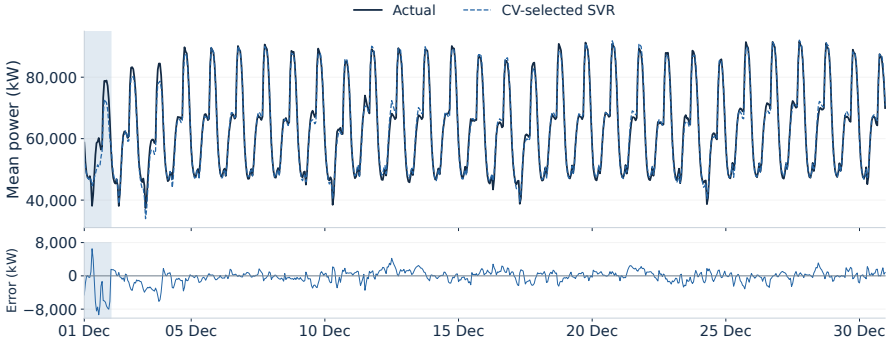
MAPE: **1.78%**

$R^2$: **0.9869**

**Residual SVR: observed test-best; not CV-selected. Absolute SVR remains final.**

December 1–30 only; results condition on observed target-hour weather.

# Good daily tracking still leaves concentrated errors




**Bias: -202.1 kW**

Error = forecast − actual.

**Worst day: 1 December (Mawlid)**

MAE 4,579.2 kW; highlighted in blue.

Possible explanation: the holiday flag may transfer larger Eid demand reductions to Mawlid.

# Compared with published Tetouan studies

**VALUES AS REPORTED BY EACH STUDY, NOT RE-RUN**

| Study | Target | Forecast horizon | Split | MAPE | $R^2$ |
|---|---|---|---|---|---|
| RoSe et al. (2023) | Total | none (no past load) | 70/30, order not stated | – | 0.976 |
| Islam et al. (2025) | not stated | 10 min ahead | 75/15/10 chronological | 6.3% | 0.853 |
| Alsalem (2025) | Zone 3 only | 10 min ahead | 70/30, order kept | – | 0.989 |
| Biswas et al. (2026) | mean of 3 zones | not stated | chronological | 1.04% | – |
| **This study** | **Total** | **24 h (day-ahead)** | **chronological** | **1.78%** | **0.987** |

**Longer horizon:** others forecast 10 minutes ahead, use no past load, or state no horizon.

**Strict evaluation:** model chosen on chronological CV; December used only once.

**Named reference:** −50.7% RMSE versus copying yesterday (Naive-24h).

# Findings & limitations

## FINDINGS

1. Choose the final model on chronological CV; report December separately.
2. Daily lags give the largest RF gain; weekly/rolling features hurt.
3. The CV-selected SVR halves December Naive-24h error ($-50.7\%$).

## LIMITATIONS

- Weather is observed at the target hour; a real midnight forecast needs weather forecasts.
- Prompt recent-load measurements assumed.
- One city, one year, one 30-day test.

**Next: issuance-time weather and independent seasonal tests.**

# Conclusion & Q&A

**CV-SELECTED ABSOLUTE-TARGET SVR / DECEMBER 2017**

**1,624.6 kW**
RMSE on 720 hourly observations

**50.7%**
lower RMSE than Naive-24h

**A promising chronological benchmark, conditional on observed weather.**

## Thank you. Questions?